משוואות & פיזיקה מתמטית

מועד ב' [illegible]

1. א.

$$XZ\frac{\partial z}{\partial x}+Yz\frac{\partial z}{\partial Y}=Xe^{X+Y}$$

נחפש פת' סתומה w

$$Xz\frac{\partial w}{\partial X}+Yz\frac{\partial w}{\partial Y}+Xe^{X+Y}\frac{\partial w}{\partial z}=0$$

$$\frac{dX}{Xz}=\frac{dY}{Yz}=\frac{dz}{Xe^{X+Y}}$$

$$\frac{dX}{Xz}=\frac{dY}{Yz}$$

$$\frac{dX}{X}=\frac{dY}{Y}$$

$$\ln X=\ln Y+\tilde{C}_1$$

$$\frac{X}{Y}=C_1 \Rightarrow X=C_1Y$$

$$\frac{dY}{Yz}=\frac{dz}{Xe^{X+Y}}$$

$$\frac{dY}{Yz}=\frac{dz}{C_1Ye^{Y(1+C_1)}}$$ נציב את C_1

$$\frac{dY}{Y}=\frac{zdz}{C_1Ye^{C_1Y+Y}}$$

נבצע החלפה $U=z^2$

$dU=2zdz$

$zdz=\frac{1}{2}dU$

$$\frac{dY}{Y}=\frac{dU}{2C_1Ye^{Y(1+C_1)}}$$

$$\frac{dU}{dY}=2C_1e^{Y(1+C_1)}$$

$Y(1+C_1)$ $\quad$ $Y(1+C_1)$

$$z^2 = \frac{2C_1}{1+C_1} e^{y(1+C_1)} + C_2$$

$$C_2 = z^2 - \frac{2C_1}{1+C_1} e^{y(1+C_1)}$$

$W(C_1,C_2)=0$ ω פונ' סתמית

אז $C_2 = \varphi(C_1)$

$$z^2 = \frac{2x}{y(1+\frac{x}{y})} e^{y(1+\frac{x}{y})} + \varphi(\frac{x}{y})$$

$$z^2 = \frac{2x}{y+x} e^{y+x} + \varphi(\frac{x}{y})$$

עקום $= \{ z^2 = 2e^{x+1} , \quad y=1 \}$?

$2e^{x+1} = \frac{2x}{x+1} e^{x+1} + \varphi(x)$ על העקום

$$\varphi(x) = 2e^{x+1}\left(1-\frac{x}{x+1}\right)$$

$z^2 = \frac{2x}{x+y} e^{\frac{x}{y}+1} + 2e^{\frac{x}{y}+1}\left(1-\frac{x}{x+y}\right)$ בסך הכל:

$L_x U$ (תגובה לגבי x)

$L_y G$ (הכללית לגבי y) $-\infty < x < \infty$

$x_0 \rightarrow x$
$x \rightarrow y$ (מקור הכלל)

כל פונ' גרין מקיימת

$$L_y G(x,y) = -\delta(y-x)$$

$$\lim_{y\to\pm\infty} G(x,y) = \alpha_\pm e^{\pm iky}$$

נקבל את הגבול ע"י נגזר x כאשר y משתנה

$$-\infty \underset{V_1(y)}{\text{———}} \overset{|}{\underset{x}{}} \underset{V_2(y)}{\text{———}} \infty$$

~~$G = \frac{1}{-Wp}\begin{cases} V_1(x)V_2(x) \\ V_2(y)V_1(x) \end{cases}$~~

עבור פונ' גרין חד-מימדית יהיה כך:

$$G = -\frac{1}{Wp}\begin{cases} V_1(y)V_2(x) & -\infty < y < x \\ V_2(y)V_1(x) & x < y < \infty \end{cases}$$

נחזור לגבול:

$$\lim_{y\to\infty} G(x,y) = e^{iky}\alpha_+$$

מצד שני

$$\lim_{y\to\infty} G(x,y) = V_2(y)\left(\frac{1}{Wp}\right)V_1(x)$$

$$\Downarrow$$

$$V_2(y) = e^{iky}$$

$$\alpha_+ = \frac{1}{Wp}V_1(x)$$

באופן דומה:

$$V_1(y) = e^{-iky} \Longleftarrow \lim_{y\to-\infty} G(x,y) = e^{-iky}\alpha_-$$

$$V_1(Y)=e^{-ikY}$$

$$V_2(Y)=e^{ikY}$$

$$W=\begin{vmatrix} e^{-iky} & e^{iky} \\ -ike^{-iky} & ike^{iky} \end{vmatrix} = ike^{-iky+iky}+ike^{-iky+iky}=2ik$$

$$\alpha_+=\frac{1}{2ik}e^{-ikx}$$

$$\alpha_-=-\frac{1}{2ik}e^{ikx}$$

$$G=-\frac{1}{2ik}\times\begin{cases} e^{ikx-iky} & -\infty<y<x \\ e^{-ikx+iky} & x<y<\infty \end{cases}$$

$$G(x,y)=-\frac{1}{2ik}e^{ik|y-x|}$$

$$\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2}+\psi_{(x)}V(x)=E\psi$$

[illegible] ?

$$\frac{d^2\psi}{dx^2}+\frac{2mV}{\hbar^2}\psi_{(x)}=\frac{2mE}{\hbar^2}\psi$$

[illegible]

$$\frac{d^2\psi}{dx^2}+\frac{2m(E-V)}{\hbar^2}\psi=0$$

$$k_1^2=\frac{\hbar^2k^2}{2m}\cdot\frac{2m}{\hbar^2}=k^2 \Leftarrow E=\frac{\hbar^2k^2}{2m} \Leftarrow k^2=\frac{2mE}{\hbar^2}$$

$$\frac{d^2\psi}{dx^2}+\frac{2mE}{\hbar}\psi_{(x)}=\frac{2m}{\hbar^2}V_{(x)}\psi_{(x)}$$

$$\tilde{V}(x)=\frac{2m}{\hbar^2}V_{(x)}$$

$$\frac{d^2\psi}{dx^2}+k_1^2\psi=\tilde{V}_{(x)}\psi_{(x)}$$

[illegible]

$$f(x) = \lambda \int_{-\pi}^{\pi} \left(x\cos y + \frac{3}{2\pi^2} y\cos x + \cos x \cos y - \frac{3}{2\pi^2} xy \right) f(y)\,dy$$

$$f(x) = \lambda x \underbrace{\int_{-\pi}^{\pi} \cos y\, f(y)\,dy}_{A} + \lambda \cos x \frac{3}{2\pi^2} \underbrace{\int_{-\pi}^{\pi} y f(y)\,dy}_{B} + \lambda \cos x \underbrace{\int_{-\pi}^{\pi} \cos y\, f(y)\,dy}_{A} -$$

$$- \frac{3\lambda x}{2\pi^2} \underbrace{\int_{-\pi}^{\pi} y f(y)\,dy}_{B}$$

$$A = \int_{-\pi}^{\pi} \cos y \left(\lambda y A + \lambda \cos y \frac{3}{2\pi^2} B + \lambda \cos y\, A - \frac{3\lambda y}{2\pi^2} B \right) dy$$

$$B = \int_{-\pi}^{\pi} y \left(\lambda y A + \lambda \cos y \frac{3}{2\pi^2} B + \lambda \cos y\, A - \frac{3\lambda y}{2\pi^2} B \right) dy$$

$$A = \lambda A \int_{-\pi}^{\pi} y\cos y\,dy + \lambda B \frac{3}{2\pi^2} \int_{-\pi}^{\pi} \cos^2 y\,dy + \lambda A \int_{-\pi}^{\pi} \cos^2 y\,dy - \frac{3\lambda B}{2\pi^2} \underset{0}{\cancel{\int_{-\pi}^{\pi} y\cos y\,dy}}$$

$$B = \lambda A \int_{-\pi}^{\pi} y^2\,dy + \frac{3\lambda B}{2\pi^2} \int_{-\pi}^{\pi} y\cos y\,dy + \lambda A \int_{-\pi}^{\pi} \cos^2 y\,dy - \frac{3\lambda B}{2\pi^2} \int_{-\pi}^{\pi} y\cos y\,dy \searrow 0$$

$$\int_{-\pi}^{\pi} y^2\,dy = \left.\frac{y^3}{3}\right|_{-\pi}^{\pi} = \frac{2}{3}\pi^3$$

$$\int_{-\pi}^{\pi} y\cos y\,dy = \underset{\searrow 0}{\left. y\sin y \right|_{-\pi}^{\pi}} - \int_{-\pi}^{\pi} \sin y\,dy = \left.\cos y\right|_{-\pi}^{\pi} = \underset{\substack{\downarrow \\ \text{NO SENON} \\ \cos \pi}}{\cos\pi - \cos(-\pi)} = \boxed{0}$$

$$\int_{-\pi}^{\pi} \cos^2 y\,dy = \int_{-\pi}^{\pi} \left(\frac{1}{2} + \frac{1}{2}\cos 2y \right) dy = \frac{\pi}{2} + \frac{\pi}{2} + \frac{1}{4}\big(\underset{\searrow 0}{\sin 2\pi} - \underset{\searrow 0}{\sin(-2\pi)} \big) = \pi$$

$$A = \frac{3\lambda B}{2\pi} + \lambda\pi A \quad \Rightarrow A(1 - \lambda\pi) = \frac{3\lambda B}{2\pi}$$

~~$A(1-\lambda\pi) - \frac{3\lambda}{2\pi}B = 0$~~

~~$A\left(\frac{2\pi^3}{3}\lambda\right) - B(1+\lambda\pi) = 0$~~

~~$\begin{vmatrix} 1-\lambda\pi & -\frac{3\lambda}{2\pi} \\ \frac{2\pi^3}{3}\lambda & 1+\lambda\pi \end{vmatrix} = 0$~~

~~$(1-\lambda\pi)(1+\lambda\pi) + \frac{3\lambda}{2\pi}\frac{2\pi^3\lambda}{3} = 0$~~

~~$1 - \pi^2\lambda^2 + \pi^2\lambda^2 = 0$~~

$$\begin{cases} A = \frac{3\lambda}{2\pi}B + \lambda\pi A \\ B = \frac{2}{3}\pi^3\lambda A - \lambda\pi B \end{cases} \qquad \begin{vmatrix} \lambda\pi-1 & \frac{3\lambda}{2\pi} \\ \frac{2}{3}\pi^3\lambda & -\lambda\pi-1 \end{vmatrix} = 0$$

$$-(\lambda\pi-1)(\lambda\pi+1) - \pi^2\lambda^2 = 0$$

$$\lambda^2\pi^2 - 1 + \pi^2\lambda^2 = 0$$

$$2\pi^2\lambda^2 = 1$$

$$\lambda^2 = \frac{1}{2\pi^2}$$

$$\lambda = \pm\frac{1}{\sqrt{2}}\frac{1}{\pi}$$

יש הצבה נוספת. [illegible] [illegible]